

BIOL 413:

PHAGE DISCOVERY LAB

Fall 2021

Biotechnology Program, Department of Life Sciences
University of New Hampshire, Manchester, NH 03101

WHO*

Instructor: Dr. Kristen Johnson

Office: Room 630

Office Hours by Zoom: Wed 10am-12pm; *and by appointment*

WHERE/WHEN

Lab (RM 560): Tu/Th, 9:30-11:30 am

FEATURING · 3 ONLINE QUIZZES (15 PTS) · MATH
PACKET (10 PTS) · 2 LAB MEETING PRESENTATIONS
(10 PTS) · *POSTER PRESENTATION (20 PTS)* · *JOURNAL
CLUB (10 PTS)* · *ATTENDANCE, PARTICIPATION, AND
RESEARCH HABITS OF MIND (20 PTS)* · *LAB NOTEBOOK
AND BENCHMARKS (40 PTS)* ·

REQUIRED MATERIALS

TEXTBOOK

Online access to:
seaphages.org;
phagesdb.org

COURSE WEBPAGE

**Pay attention to the
Announcements section!**

Modules will contain
course materials: including
the syllabus, Power Point
presentations, handouts,
links to online quizzes,
useful websites, and
homework assignments.

OTHER STUFF

Lab Coat
Safety Goggles/Glasses

*CONTACT INFO

kristen.johnson@unh.edu
603-641-4177 (office)

THE DO's and DON'Ts...The Fine Print

ATTENDANCE:

- o Please arrive on time. Do not leave early.
- o Regular attendance is vital to your success in this course! While occasional circumstances, over which you have no control, may necessitate absence, the content presented in the activities missed by you is a segment of a broad base of information being taught. For this reason, your attendance is expected.
- o Attendance will be noted during every lab section. Excessive lab absences (two or more) from the laboratory will result in a reduction in your lab grade.
- o It is your responsibility to get all lab material from a classmate to make up what you have missed during any absence. Also, be sure to check MyCourses/Canvas for relevant Power Points, handouts, assignments, etc. In most cases, you will need to come in outside of scheduled lab time to make up missed lab hours.

TECHNOLOGY USE:

- o *Cell phones:* Please silence your cell phone before you enter the lab classroom. For this lab, in many cases, you will want to use your phone to document your results. Thus, you may bring it into the lab, but keep it in a secure (CLEAN) location, until you are ready to take the pictures – i.e, you don't want to spill bacteria on it! Thus, keeping it in a ziplock bag is a good idea.
- o *Computer Use:* You may bring your laptop into the lab to document your lab progress in your online lab notebook. Be sure that you are careful to have it off to the side of your working area and that you wipe down computer surfaces prior to leaving the lab.

LAB CONDUCT:

- o *Food:* There is a strict policy of **NO** food or drinks in the laboratory.
- o You must wear close-toed shoes, long pants (or long skirt), a lab coat and protective eye wear/goggles at **all times** while we are doing experiments in the lab.

Please review these [Important University Policies for Students](#). Reach out to your instructor with any questions.

Methods of Student Evaluation: The laboratory grade will be based on the student's performance on the following:

Method of Evaluation	Point Value
Homework: Math Packets I and II	10
Online Lab Notebook and Demonstration of Benchmarks*	40
Journal Club Article Online Annotations, Questions, and Discussion (5 pts ea)	10
3 Online Quizzes (5 pts ea)	15
Attendance, Lab Citizenship & Research Habits of Mind	20
Lab Meeting Presentations (5 pts ea)	10
Final Poster Presentation	20
Total Points Possible	125

***Benchmarks:** *More information on these benchmarks and the requirements for the lab notebook are included in a separate document.*

1. Plaque Assay Plate
2. Spot Titre and Full Titre Plate
3. High Titre Phage Lysate (w/Calculations)
4. DNA Concentration & Restriction Digest Analysis
5. TEM Picture
6. Complete phage data entered into phages.db and phage archived
- * 7. Additional testing: Host range, lysogeny, etc. (if time)

BIOL 413 PHAGE DISCOVERY: LAB SCHEDULE

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Schedule subject to change depending upon experiment outcomes.

Week/ Date	Lab Activities	<u>Pre-Lab</u> Reading/Viewing	Quizzes	Report/Assignment Due
1 8/31, 9/1	Introduction to SEA-PHAGES Intro to lab work, aseptic technique, and lab safety (stations). Sample Collection.	Check out seaphages.org Tu: Read Section 1, 2; Th: Read Section 5-intro, 5.1		Soil samples due (9/1)
2 9/7, 9	Phage Isolation	Tu: Read Sections 4, 5.2, 5.3; Watch Videos: Direct Isolation, Plaque Assay Th: Read Sections 3, 5.5; Watch Video: Enrichment Protocol		Math Packet I Due: Basic Calculations and Dilutions
3 9/14, 16		Tu: Read Sections 5.4, 5.6; Watch Videos: spot test, picking a plaque Th: Read Sections 6-intro, 6.1, 6.2; Watch Videos: serial dilutions, plaque assay for purification		Math Packet II Due: Plaque forming units calculations and beyond
4 9/21, 23	Phage Purification	Tu: Read Sections 6.3, 6.4, 6.5	Online Quiz 1	Lab Meeting I
5 9/28, 30		Tu: Read Section 7-intro		
6 10/5, 7	Phage Amplification High Titer Lysate (Hard Deadline)	Tu: Read Sections 7.1, 7.2		Journal Club 1
7 10/12, 14			Online Quiz 2	Lab Meeting II
8 10/19, 21				
9 10/26, 28	DNA Extraction and Restriction Analysis	Tu: Read Section 9-all		Lab Meeting III
10 11/4		Tu: Read Section 10-all	Online Quiz 3	
11 11/9	Microscopy	Tu: Read Section 8		Lab Meeting IV
12 11/16, 18	DNA and Microscopy Analysis *DNA Submission DEADLINE*			Journal Club 2
13 11/23	<i>Work on Poster</i>			
14 11/30, 12/2	<i>Finish Poster; Practice Poster Presentation</i>			Posters Due (11/30); Winter Symposium (12/2)
15 12/7, 9	Lab Clean-Up and Phage Archival	Tu: Read Section 7.3		